

Also, current technological obstacles that are apparent in plastic electronics were not detrimental to the commercialisation of the product."

Sensor to monitor head impacts in sports. In the last one year or so, the scenario of flexible electronics has rapidly changed. "Main driving applications in this field are related to flexible, rollable, foldable and paperlike displays for the consumer electronics industry," notes Petti.

Wearable applications with flexible electronics are envisioned not only in the huge display industry but also in the healthcare sector. Here, MC10 is shaping the field with its ultra-thin flexible skin sensor patch, Biostamp.

An interface between human and electronics, Biostamp is a seamless sensing soft sticker, capable of stretching, flexing and moving with the body. Powered by thin-film battery technology, this sensor can measure a variety of physiological functions such as data from the heart, muscles, brain, body temperature and body movement.

Biostamp has been used commercially in Reebok's CHECKLIGHT, a head-impact indicator that uses a multiple of these sensors to capture head-impact data during play when athletes are unaware of the severity of a blow to the head.

Lighting system that uses flexible circuitry. Cohda, a UK based design studio, has worked with CPI on the integration of printed electronics to significantly enhance the functionality of their wireless Crypsis Lighting product, developing the product from prototype to full commercial manufacture. Crypsis Lighting offers wireless ultra-bright LEDs that can be repositioned and dimmed within a transparent glass panel using an external magnetic control puck. The system is a fully-interactive low-voltage lighting unit and is currently being used within a diverse range of products and markets such as interior design, exhibition design, museum, retail,

Technologies enabling commercial flexible products today

The technologies currently available commercially include flexible thin-film transistors and logic circuits (with amorphous silicon, organic, metal-oxide, low-temperature polycrystalline silicon semiconducting materials), flexible organic light emitting diode (OLED) based displays, flexible memories and batteries, as well as flexible photovoltaics and sensors.

Additionally, many types of flexible and transparent conductors are also available commercially, ranging from indium-tin-oxide (ITO) to nanotubes and nanowires. All these materials and technologies are mature, and the devices based on these are commercially available and ready to be integrated into more complex flexible systems that can enable a wide range of applications.

—*Luisa Petti, PhD student, Wearable Computing Laboratory, Swiss Federal Institute of Technology, Zurich*

architecture and contemporary lighting.

Earlier, Crypsis Lighting utilised silicon based electronics for their lighting units. Bagshaw says, "Due to the rigid nature of the circuitry, Cohda encountered a number of issues in research and development including the transfer of power to the light units and voltage drop with the electronics within the light units."

He adds, "CPI worked with Cohda to use printed electronics to bring flexibility and conformability into the design of their light unit device. These properties enabled the electronics to conform to the surface contact of the glass, resulting in the elimination of voltage drop and an increase in conductivity levels."

Pure-copper-conductive ink for the wearable world.

In November 2014, DuPont Microcircuit Materials (DuPont) introduced their PE510 photonic copper product. PE510 is a cost-effective alternative to silver-conductor inks for a variety of possible applications, and is the newest product in a suite of conductive ink materials specifically tailored for use in certain types of antennae, membrane touch switches (MTSes), radio frequency identification (RFID) and consumer electronics applications.



A player wearing Reebok Checklight, a skullcap with sensors to monitor head impacts during play (Image courtesy: thechronicleherald.ca)



CPI worked with Cohda to use printed electronics to bring flexibility and conformability into the design of their wireless device Crypsis Lighting (Image courtesy: www.uk-cpi.com)

Stan Farnsworth, VP - marketing, NovaCentrix, says, "This electrically-conductive ink is designed for use on polymeric substrates and reaches optimal performance when processed with PulseForge photonic curing tool from NovaCentrix." These inks provide designers with higher flexibility to design antennae, enabling a lower total manufacturing cost, while meeting electrical performance requirements.